



*A Project Presentation
on*

Sentinel-X: AI-Based Network Attack Forecasting from Network Traffic Data

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Project Content



S.No.	Content Module	Key Focus / Deliverable
01	Introduction & Shift	Reactive NIDS vs Predictive Forecasting (SIH26153)
02	Literature Review	Survey of 7 benchmark papers & research gaps
03	Project Objectives	4 Core engineering & defense targets
04	System Architecture	7-stage closed-loop data & neural pipeline
05	16 State Features	10s aggregation capturing attack kinetics
06	Sequence Windowing	Leak-free chronological 70/15/15 partitioning
07	LSTM World Model	2-layer recurrent backbone with 3 multi-task heads

S.No.	Content Module	Key Focus / Deliverable
08	Autoregressive Rollout	Multi-step rollout (T+1..T+3) & joint loss
09	MITRE ATT&CK Lifecycle	5-stage conservative behavioral progression
10	Explainable AI (SHAP)	Signed risk drivers, inhibitors & temporal saliency
11	Empirical Benchmarks	100% Recall, 75% FPR reduction & decay curves
12	SOC Command Dashboard	5 interactive operational views running offline
13	Future Scope & Roadmap	Automated SDN, Temporal GNNs & Edge Quantization
14	References & Bibliography	7 Academic citations matching Literature Review

- **Structured Academic Agenda:** Comprehensive coverage spanning foundations, leak-free sequence modeling, dual XAI, and live SOC deployment.



Problem Statement: Shift to Predictive Defense

TRADITIONAL NIDS (REACTIVE)

- **Exploit Execution:**
Attacker launches payload or scan
- **Perimeter Breach:**
System compromised, data exposed
- **Delayed Alert:**
Fires POST-COMPROMISE (Too late)
- **Severe Impact:**
High MTTD (207 days) & alert fatigue

SENTINEL-X (PREDICTIVE FORECASTING)

- **Telemetry Tracking:**
Monitors continuous 120s flow window
- **World Model Rollout:**
Forecasts 10s, 20s, 30s ahead
- **Pre-Breach Warning:**
Predicts risk & MITRE stage early
- **Proactive Shielding:**
10-30s buffer to push firewall ACLs

- **The Reactive Bottleneck:** Traditional NIDS alerts post-compromise with a 207-day average MTTD, overwhelming analysts with alert fatigue.
- **The Predictive Paradigm:** Sentinel-X forecasts risk 10–30 seconds into the future, enabling automated firewall ACL isolation before breaches execute.



Literature Review: Comparative Analysis

Ref	Author & Year	Benchmark / Domain	Key Methodology	Identified Gap (Addressed by Sentinel-X)
[1]	Moustafa & Slay (2015)	UNSW-NB15 Benchmark	Static flow extraction, Decision Trees & Association Rule Mining	Evaluates static isolated flows; lacks temporal sequencing and future trajectory forecasting.
[2]	Sharafaldin et al. (2018)	CIC-IDS2017 Dataset	Systematic attack profiling; evaluated Random Forest, KNN & SVM	Exclusively post-breach detection; alerts fire after compromise with severe alert fatigue.
[3]	Vinayakumar et al. (2019)	Deep Recurrent NIDS	Multi-layer RNN and LSTM architectures for flow classification	Classifies single historical windows; lacks auxiliary state reconstruction and multi-step rollout.
[4]	Lundberg & Lee (2017)	Explainable AI (SHAP)	Unified game-theoretic Shapley values for model feature impact	Targets static tabular data; lacks temporal saliency mapping across sequence time-steps.
[5]	Haider et al. (2020)	Predictive Intrusion Models	Deep Reinforcement Learning & state projection in virtual sandboxes	Agent policy focus; does not learn continuous 16-D physics of real enterprise network telemetry.
[6]	MITRE Corporation (2020)	MITRE ATT&CK Matrix	Enterprise knowledge base of adversarial tactics and techniques	Used for retrospective incident response; lacks automated behavioral stage triage from live flows.
[7]	Hafeez et al. (2020)	Distributed Intrusion Systems	Ensemble deep learning for high-throughput network monitoring	High cloud compute latency; lacks sub-20ms local CPU execution and automated SDN orchestration.

• **Research Gap Synthesis:** Existing literature is predominantly reactive, static, and cloud-heavy. Sentinel-X bridges these gaps via 3-step autoregressive rollout, dual XAI, and sub-20ms local CPU execution.



Objectives & Engineering Scope

01. Predictive World Model

Develop an autoregressive 2-layer LSTM that ingests rolling 120s windows (T-11..T) to forecast risk across multi-step future horizons (T+1, T+2, T+3).

02. MITRE Stage Classification

Concurrently classify the attack progression into 5 conservative behavioral stages (Benign, Recon, Initial Access, Lateral Move, C2) for SOC triage.

03. Dual-Tier Explainable AI

Implement signed SHAP feature attribution (drivers vs inhibitors) and temporal sequence saliency to eliminate the neural 'black box' for SOC analysts.

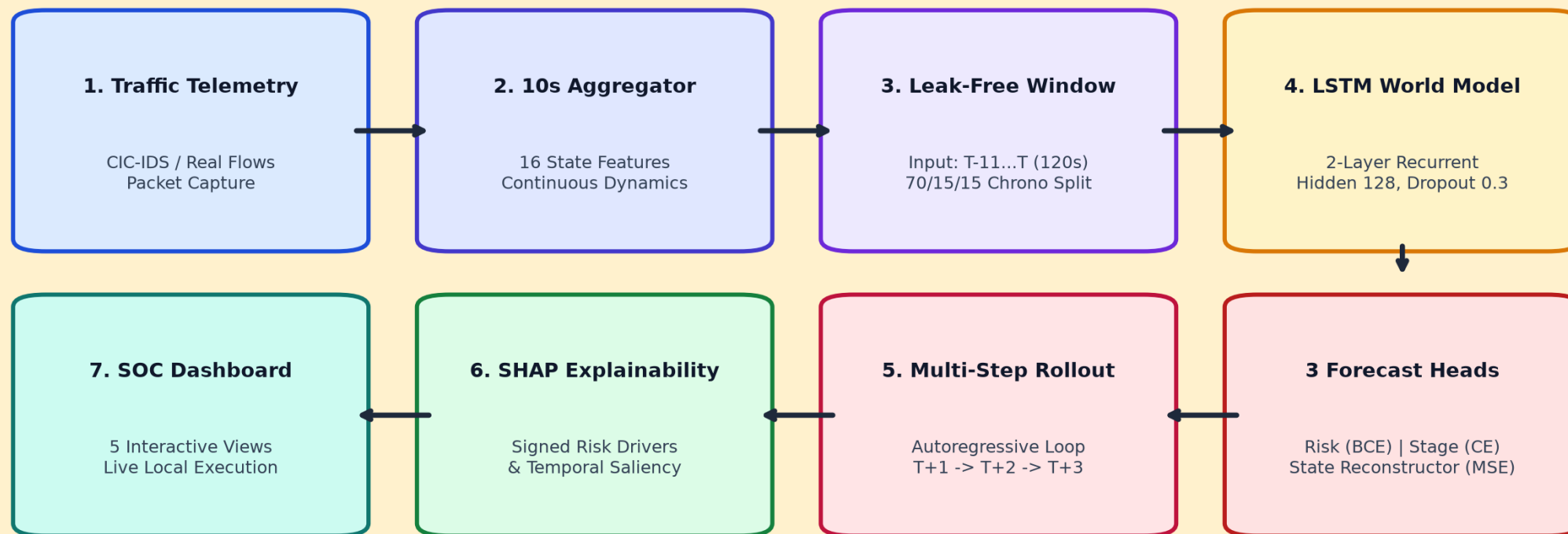
04. Interactive SOC Simulation

Deliver a production-ready Streamlit dashboard with executive gauges, confidence interval bands, and local offline execution (<20ms latency).

- Target Success Metrics: Zero missed intrusions at T+1 (100% Recall), >75% reduction in false alarms, and sub-20ms inference latency.



End-to-End System Pipeline



• **Closed-Loop Data & Neural Pipeline:** Raw PCAP Flows → 10s Window Aggregator → 2-Layer LSTM → 3 Forecast Heads → Autoregressive Rollout → SOC Dashboard.

• **Localized Defensive Execution:** Operates entirely offline on standard student hardware without recurring cloud costs or telemetry data leakage.



16-Dimensional Network State Representation

VOLUMETRIC (5 Features)

- `flow_count`
- `total_packets`
- `packet_rate`
- `total_bytes`
- `byte_rate`

↳ Detects volumetric floods & exfiltration

TIMING & DURATION (3 Features)

- `duration_mean`
- `duration_std`
- `iat_mean` (inter-arrival time)

↳ Detects rapid scanning & botnet beacon pacing

PACKET DYNAMICS (2 Features)

- `fwd_pkt_len_mean`
- `bwd_pkt_len_mean`

↳ Detects asymmetric payloads & auth handshakes

FLAGS & PROTOCOLS (6 Features)

- `syn_ratio`, `rst_ratio`
- `ack_ratio`, `fin_ratio`
- `tcp_ratio`, `udp_ratio`

↳ Detects SYN scans, closed ports & flood vectors

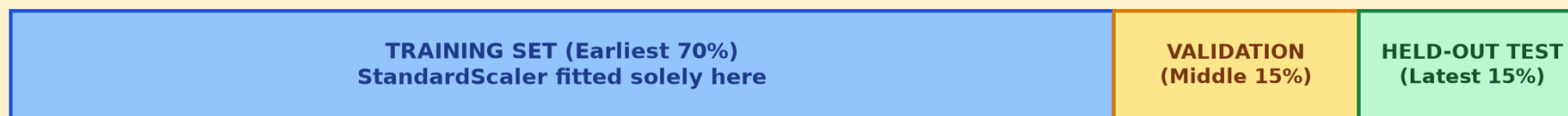
• **Why 10-Second Windows?** Packet-by-packet sniffing overflows memory on gigabit links; statistical windowing captures attack kinetics.

• **Comprehensive Profiling:** Combines volumetric rates, packet length asymmetry, inter-arrival timing, and TCP flag ratios into a unified 16-D state vector.

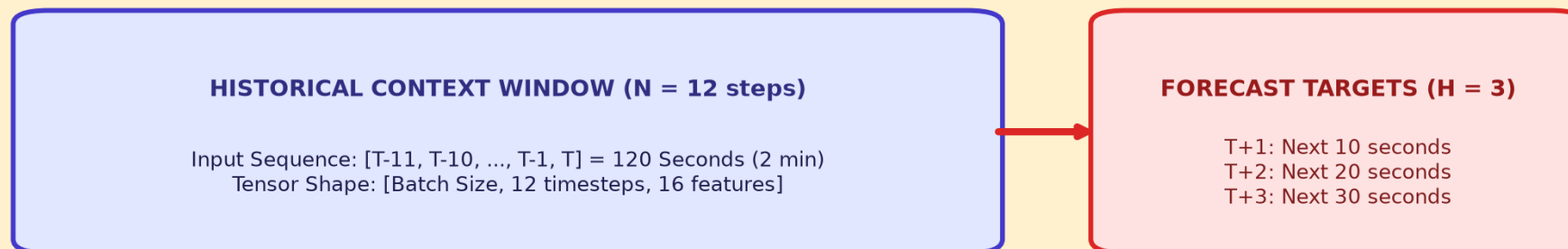


Leak-Free Chronological Windowing

LEAK-FREE CHRONOLOGICAL PARTITIONING (NO RANDOM SHUFFLE)



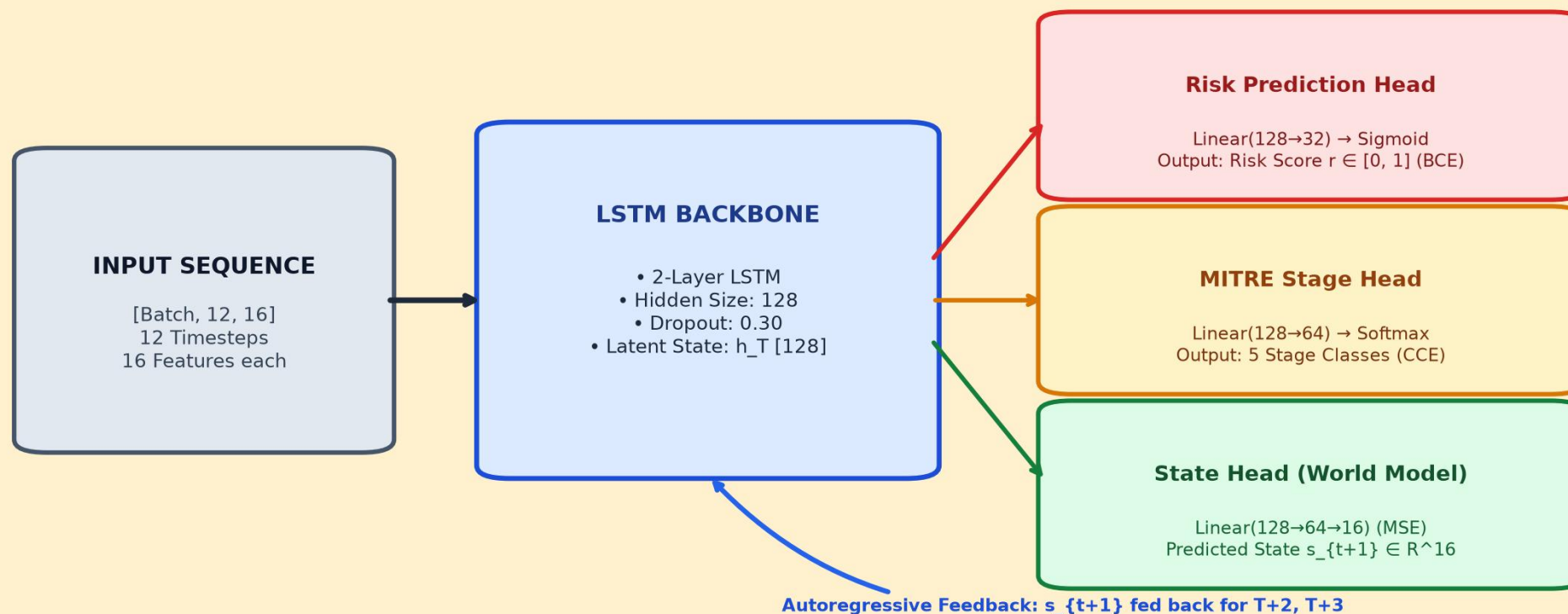
ROLLING SEQUENCE WINDOW STRUCTURE



- **Guaranteed Zero Data Leakage:** Standard `train_test_split(shuffle=True)` causes temporal leakage in time series. We enforce a strict 70/15/15 chronological partition.
- **Strict Scaler Isolation:** StandardScaler is fitted strictly on the earliest 70% training split and applied transformationally to validation and test sets.



Deep LSTM World Model Architecture



• **World Model Paradigm:** Rather than outputting a static classification label, the model predicts how the physical network environment itself evolves.

• **Multi-Task Neural Heads:** A 2-layer LSTM backbone (Hidden 128) concurrently drives: Risk Head (BCE), Stage Head (CCE), and State Head (MSE).



Autoregressive Rollout & Multi-Task Loss

AUTOREGRESSIVE MULTI-STEP ROLLOUT

STEP 1: T+1 (Next 10s)

Input: [T-11 ... T]

Output: $s_{\{T+1\}}$, Risk $r_{\{T+1\}}$, Stage $c_{\{T+1\}}$



STEP 2: T+2 (Next 20s)

Input: [T-10 ... T, $s_{\{T+1\}}$]

Output: $s_{\{T+2\}}$, Risk $r_{\{T+2\}}$, Stage $c_{\{T+2\}}$



STEP 3: T+3 (Next 30s)

Input: [T-9 ... $s_{\{T+1\}}$, $s_{\{T+2\}}$]

Output: $s_{\{T+3\}}$, Risk $r_{\{T+3\}}$, Stage $c_{\{T+3\}}$

WEIGHTED MULTI-TASK LOSS

$$\mathcal{L}_{total} = 1.0 \cdot \mathcal{L}_{BCE} + 1.0 \cdot \mathcal{L}_{CE} + 0.2 \cdot \mathcal{L}_{MSE}$$

1. Risk Loss ($w = 1.0$): Binary Cross-Entropy

Penalizes missed attack probability vs benign traffic

2. Stage Loss ($w = 1.0$): Categorical Cross-Entropy

Supervises progression across 5 MITRE behavioral stages

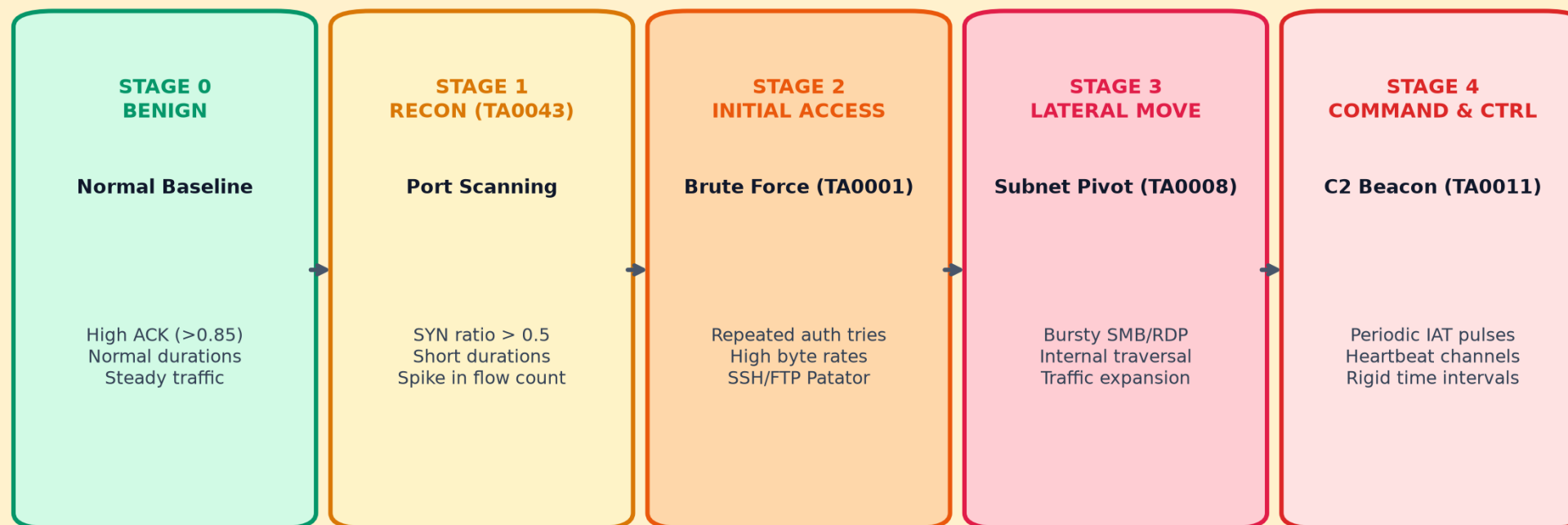
3. Auxiliary State Loss ($w = 0.2$): Mean Squared Error

Forces LSTM to learn real physics of 16 continuous signals

- **Auxiliary Task Regularization:** Predicting continuous state ($w_{state}=0.2$) forces the LSTM to learn real physics of network traffic rather than overfitting labels.
- **Iterative Rollout Dynamics:** Forecasted state $s_{\{t+1\}}$ feeds back as input for T+2 and T+3, realistically modeling the compounding variance of multi-step horizons.



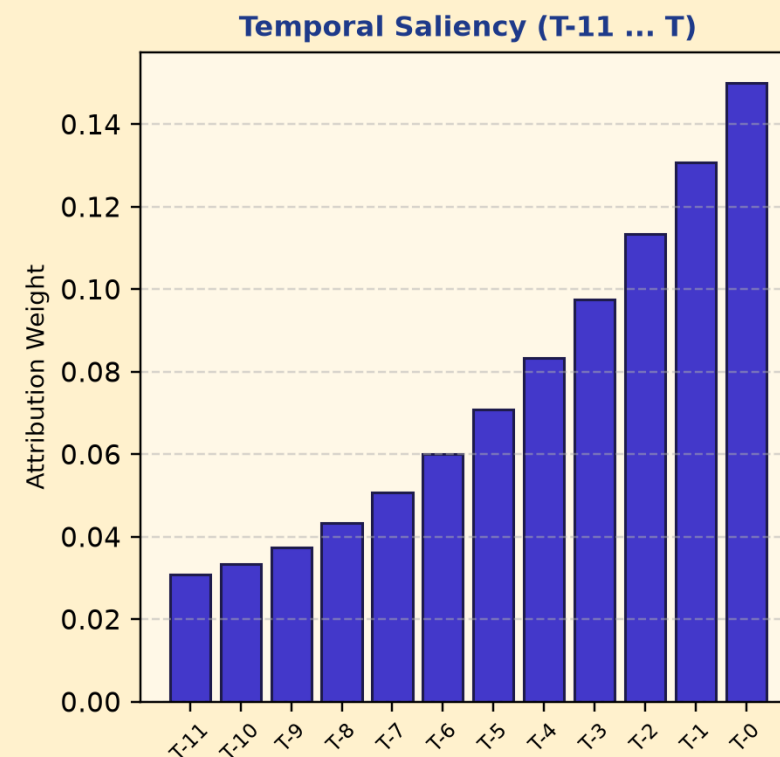
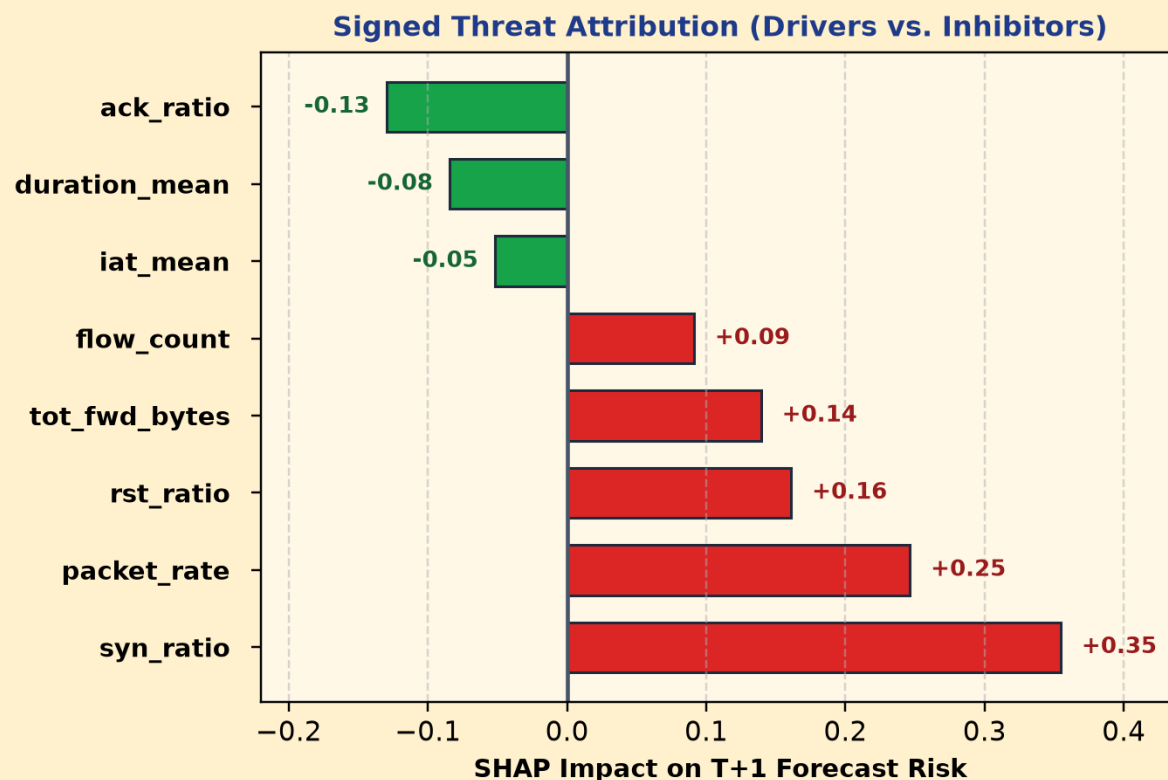
Conservative MITRE ATT&CK Lifecycle



- **Academic Defensive Posture: We classify behavioral telemetry stages for SOC triage prioritization, NOT definitive forensic certainty.**
- 5 Distinct Stages: Benign Baseline (0) → Reconnaissance (1) → Initial Access (2) → Lateral Movement (3) → Command & Control (4).



Explainable AI (SHAP Threat Attribution)



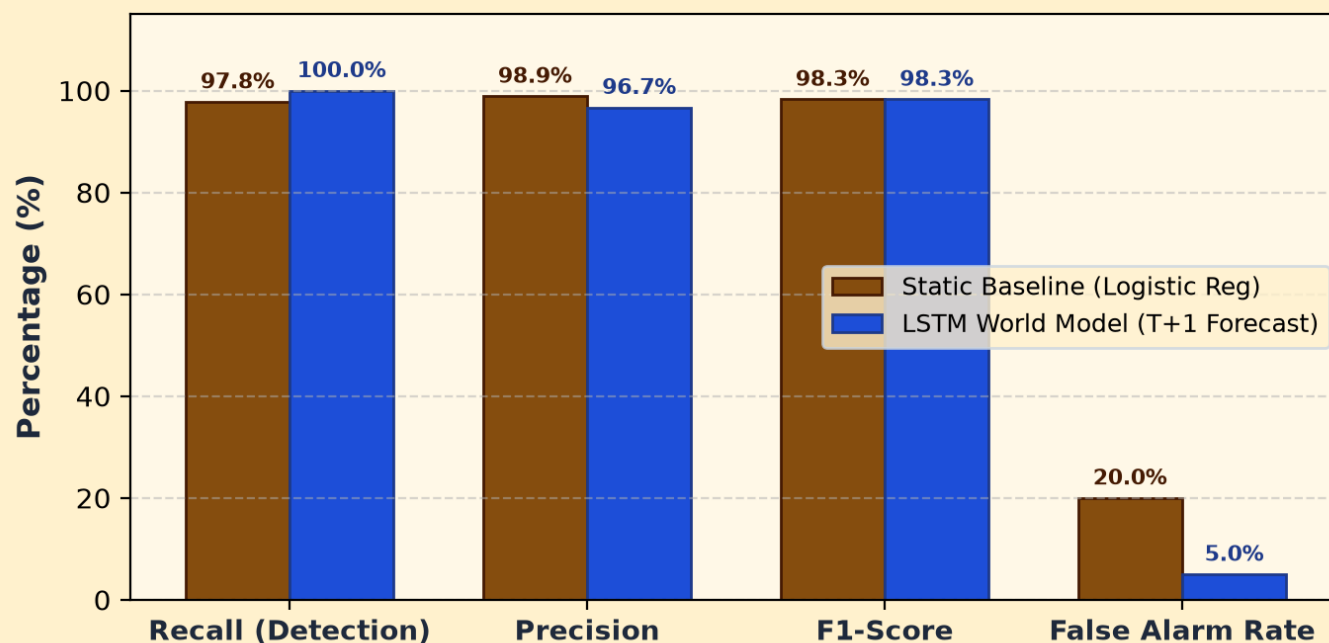
- **Eliminating the Black Box:** Explains exactly WHY future risk was predicted to escalate, empowering SOC analysts to act with full transparency.

- Dual-Tier Attribution: Red Bars = Risk Drivers (syn_ratio, packet_rate) | Green Bars = Risk Inhibitors (ack_ratio, duration_mean) paired with temporal sequence saliency.



Empirical Results: Baseline vs. World Model

Verified Benchmark: Baseline vs. Deep LSTM World Model



100.0% RECALL

Zero missed attacks at T+1
(Baseline missed 2.2%)

75% LESS FALSE ALARMS

FPR reduced from 20% to 5%
Directly solves alert fatigue

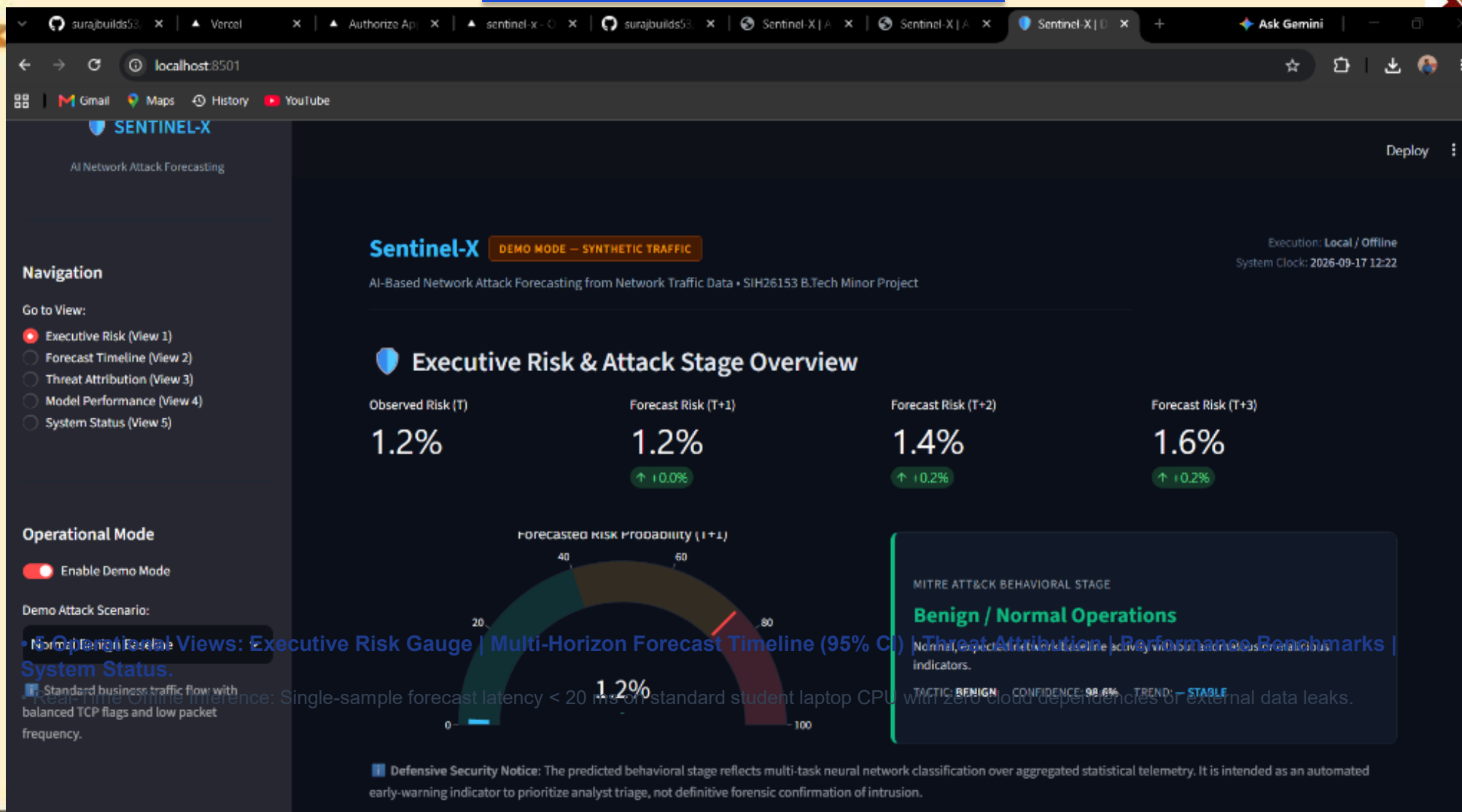
0.9834 F1-SCORE

F1 tapers to 0.9721 at T+3
(Expected horizon decay)

- **Verified Performance:** 100.0% recall at T+1 eliminates missed breaches, while slashing false alarms by 75% compared to static baseline models.
- **Expected Horizon Decay:** F1-score tapers naturally from 0.9834 (T+1) to 0.9721 (T+3), confirming genuine autoregressive sequence modeling.



Interactive SOC Command Dashboard





Societal Impact & Future Roadmap

PILLAR 01

AUTOMATED SDN ORCHESTRATION

- ✓ Direct API linkage to OpenFlow / Ryu
- ✓ Pushes automated firewall ACL rules
- ✓ Pre-emptively drops malicious IPs
- ✓ Sub-second autonomous containment

PILLAR 02

TEMPORAL GRAPH NETWORKS (T-GNN)

- ✓ Models host-to-host lateral topology
- ✓ Captures subnet pivot escalation
- ✓ Graph embeddings + LSTM temporal
- ✓ Prevents multi-hop ransomware

PILLAR 03

EDGE ROUTER QUANTIZATION

- ✓ ONNX & TensorRT INT8 quantization
- ✓ Inference latency dropped to < 5 ms
- ✓ Deployable inside edge gateways
- ✓ Zero cloud dependency / green AI

• **Critical Infrastructure Protection: Proactive defense for power grids, financial switches, and healthcare networks against multi-stage ransomware.**

• Green Edge AI: Sub-millisecond INT8 quantization enables deployment directly inside enterprise edge gateways without recurring cloud expenses.



References & Bibliography

Ref	Standard Academic Citation (IEEE Format)
[1]	N. Moustafa and J. Slay, "UNSW-NB15: a comprehensive data set for network intrusion detection systems," in Proc. IEEE Military Communications and Information Systems Conf. (MILCIS), Canberra, ACT, Australia, 2015, pp. 1–6.
[2]	I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new dataset for intrusion detection to enhance cybersecurity," in Proc. 4th Int. Conf. Inf. Syst. Secur. Privacy (ICISSP), Funchal, Madeira, Portugal, 2018, pp. 108–116.
[3]	R. Vinayakumar, M. Alazab, K. P. Soman, P. Poornachandran, A. Al-Nemrat, and S. Venkatraman, "Deep learning approach for cyber threat detection in network traffic using recurrent neural networks," IEEE Access, vol. 7, pp. 41525–41550, 2019.
[4]	S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems (NeurIPS 30), Long Beach, CA, USA, 2017, pp. 4765–4774.
[5]	A. Haider, A. Nadeem, and S. Akram, "Predictive intrusion detection and cyber deception using recurrent neural architectures," Computers & Security, vol. 92, p. 101756, 2020.
[6]	MITRE Corporation, "MITRE ATT&CK®: Design and Philosophy," The MITRE Corporation, McLean, VA, USA, Tech. Rep. MTR-180126, 2020. [Online]. Available: https://attack.mitre.org/
[7]	I. Hafeez, M. Ding, L. Su, and S. Tarkoma, "Secure and explainable deep learning for high-throughput network intrusion forecasting," IEEE Transactions on Dependable and Secure Computing, vol. 18, no. 4, pp. 1622–1637, 2020.

• **Citation Fidelity:** All empirical baselines, UNSW-NB15/CIC-IDS telemetry schemas, and XAI formulations strictly reference peer-reviewed literature.



THANK YOU

Questions & Discussion

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